IT3091 - Machine Learning
Group Assignment - Initial Submission

DataCo Smart Supply Chain - Late-Delivery Risk

Group: 2026-AI-05

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1. Track

We are following the Guided Data Track. Our group code maps to the Logistics & Supply Chain domain, so we are working with the DataCo Smart Supply Chain dataset that was provided to us. We are staying within the controlled domain rather than sourcing our own data, so the Industry Explorer bonus does not apply to us.

2. Lenses (with reasons)

Primary lens: Late-delivery risk.

We picked this because the dataset already gives us a ready-made target column, Late_delivery_risk, which is a clean 0/1 flag (1 = the order was late, 0 = it was not). That means we do not have to invent or derive our own label, so the task is well-defined from day one. It also turns naturally into a binary classification problem, which lets us do a proper baseline-plus-alternatives comparison and use real evaluation metrics. Most importantly it answers an actual business question a logistics company cares about: can we flag a risky order early, before it ships, so operations can step in (upgrade the shipping mode, warn the customer, or re-route it)?

Secondary lens (optional): Delay pattern analysis.

We are not building a separate pipeline for this. We are using it as the 'why' layer on top of the primary lens. Once our model can predict risk, we look at feature importance and simple EDA breakdowns (delay rates by Shipping Mode, Market, Order Region, and product category) to explain what actually drives lateness. This keeps the two lenses connected instead of being two disconnected analyses, and it makes our final recommendation specific ('expedite Standard Class orders going to Region X') rather than generic.

We deliberately did not take a third lens (shipment priority profiling), because it overlaps heavily with late-delivery risk and would add work without adding coherence.

3. Proposed Task / Output

Task type: Supervised binary classification.

Unit of analysis: one order (one row in the dataset).

For every order, using only information that is known at the time the order is placed, predict whether that order is at risk of being delivered late.

Input (features we plan to use - all known at order time):

- Type (transaction type)
- Days for shipment (scheduled)
- Shipping Mode (Standard / First / Second / Same Day)
- Market and Order Region (where it is being delivered)
- Category Name / Department (what is being ordered)
- Customer Segment (Consumer / Corporate / Home Office)
- Order Item Quantity, order date (DateOrders), and order priority-type fields

Target / output:

- Late_delivery_risk - 1 (likely late) or 0 (likely on time)

Leakage we are deliberately excluding:

We inspected the data and found that Late_delivery_risk is simply whether the actual shipping days exceeded the scheduled days (real > scheduled → 1). In other words, a few columns are the answer in disguise. If we left them in, the model would score near-100% for the wrong reason and the whole exercise would be meaningless. We drop these and record the reason in the decision log:

- Days for shipping (real) : the actual shipping duration, only known after the order ships; the label is derived from it.
- Delivery Status : the same outcome written out in words (Advance / On time / Late / Canceled).
- Shipping date (DateOrders) : the actual dispatch date, also post-order information.

Other columns we exclude (data quality):

We also drop columns that carry no predictive signal for delivery timing: identifiers and personal data (customer name, email, masked password, customer/order IDs, street, zipcode) and empty or non-informative fields (Product Description, Product Image URL, Order Zipcode). These are noted in the preprocessing log.

Scope decision - canceled orders:

Some rows are canceled or suspected-fraud orders (e.g. Order Status = CANCELED / SUSPECTED_FRAUD). These never actually ship, so a late-delivery prediction does not apply to them. We filter them out and only model orders that genuinely ship; a cleaner, more defensible problem definition.

Unstructured (clickstream) data - excluded, with reason:

The dataset also includes an unstructured clickstream file (tokenized_access_logs.csv). We evaluated it and chose not to use it: it contains no Customer Id or Order Id, so it cannot be joined to the order records, and browsing behavior is not related to whether an order ships on time. It answers a different business question (customer browsing behavior), which is outside our chosen decision lens. Documenting this choice is itself part of our data-understanding reasoning.

4. Workflow

Our workflow runs from the business problem through to a recommendation. Each step feeds the next, and every major decision gets written into the decision log as we go.

- Step 1 - Data understanding: read every column using the data dictionary, note variable types, and mark which columns are leakage risks or non-predictive.
- Step 2 - EDA: confirm the class balance of late vs on-time, look at delay rates by Shipping Mode / Market / Region / category, and find missing values, duplicates, and outliers.

- Step 3 - Preprocessing: drop leakage and non-predictive columns, filter out canceled/fraud orders, handle missing values, encode categorical columns, and scale features for distance-based models.
- Step 4 - Modelling: train a simple baseline first, then at least three alternative models.
- Step 5 - Evaluation: compare models on precision, recall, F1, and ROC-AUC (not just accuracy), use cross-validation, and re-check for leakage.
- Step 6 - Recommendation: tie the results back to the secondary lens and give DataCo an evidence-based, practical recommendation with its limitations.

Planned models: Logistic Regression (baseline), Random Forest, and Gradient Boosting (e.g. XGBoost), with hyperparameter tuning on the strongest candidate, giving us the baseline-plus-three-alternatives the rubric asks for.

(A workflow diagram version of these steps will be included in the final report.)

5. Core Responsibility of Each Member

We split the work so that every part of the pipeline has a clear owner, while the workflow diagram and decision log are shared , everyone records their own decisions and Member 1 assembles the final diagram.

Member	IT Number	Focus area	Core responsibility
Bolonne B.R.M.	IT24102050	Framing + EDA	Problem framing canvas (stakeholder, lens justification, task/output), full EDA, and the EDA insight log.
Kodagoda D.S.	IT24101613	Preprocessing + baseline model	Cleaning, dropping leakage/non-predictive columns, filtering canceled orders, encoding, baseline Logistic Regression, and the preprocessing/feature log.
Jayasekara C.D.	IT24103494	Advanced models + tuning	Random Forest and Gradient Boosting models, plus hyperparameter tuning on the strongest candidate, documented in the decision log.
Liyanage J.L.K.L.	IT24101927	Evaluation + recommendation	Metric comparison across all models, cross-validation, final recommendation with limitations, and the reproducibility / AI-use declaration.